



PharmaVerse

Pharma Data Intelligence Platform

Complete Developer & User Guide

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React 18 • D3.js • Recharts • Claude AI • PubMed • ChEMBL • openFDA • ClinicalTrials.gov

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1. What is PharmaVerse?

PharmaVerse is an integrated pharma data intelligence platform that combines 4 specialized tools and a RAG (Retrieval-Augmented Generation) chatbot into a single web application. It demonstrates full-stack data engineering skills across the pharmaceutical domain.

1.1 Platform Architecture

The platform follows a single-page application (SPA) architecture:

- **Frontend:** React 18 with React Router v6 for client-side routing
- **Visualization:** Recharts for charts, D3.js for force-directed graphs
- **AI:** Claude API (Sonnet 4) for natural language processing
- **Data:** Live APIs (PubMed, openFDA) + embedded datasets (ChEMBL, ClinicalTrials.gov)
- **Build:** Vite 5 for fast development and optimized production builds

1.2 The 4 Tools + Chatbot

Tool	Purpose	Data Source	Key Tech
RAG Chatbot	Ask about any medicine/disease	PubMed + openFDA (live)	Claude AI + MCP
TrialLens	Clinical trials intelligence	ClinicalTrials.gov (embedded)	Recharts
DrugNexus	Drug-target knowledge graph	ChEMBL (embedded)	D3.js force graph
BioInsight	Biomedical literature search	PubMed (live + embedded)	Claude AI + MCP
FDAForge	FDA approval pipeline	openFDA (live + embedded)	ETL + Recharts

2. How to Set Up & Run

This section walks you through getting PharmaVerse running on your computer. You need: Node.js (v18+), npm, and a code editor (VS Code recommended).

2.1 Prerequisites

- **Node.js:** Download from nodejs.org — install the LTS version (18 or higher)
- **npm:** Comes bundled with Node.js automatically
- **VS Code:** Download from code.visualstudio.com (optional but recommended)
- **Git:** Download from git-scm.com (needed for GitHub push/pull)

To verify Node.js is installed, open terminal and run:

```
node --version    # should show v18.x.x or higher
npm --version     # should show 9.x.x or higher
```

2.2 Running Locally

Step 1: Clone the repository (or unzip the downloaded folder)

```
git clone https://github.com/pellurukushalkumarreddy-png/PharmaVerse.git
cd PharmaVerse
```

Step 2: Install dependencies (downloads all packages — only needed once)

```
npm install
```

This creates a node_modules folder (~250MB) with React, D3, Recharts, Vite, etc. Takes 1-3 minutes. You only do this ONCE.

Step 3: Start the development server

```
npm run dev
```

Your browser opens at <http://localhost:3000>. Hot-reload enabled — save a file and the browser updates instantly.

Step 4: Stop the server

```
Press Ctrl + C in the terminal
```

2.3 What Each npm Command Does

Command	What It Does	When to Use
npm install	Downloads all packages to node_modules/	Once after cloning
npm run dev	Starts Vite dev server (hot reload)	Every time you develop/test
npm run build	Creates production bundle in dist/	Before deploying

npm run preview	Preview the production build	To test before deploying
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3. Project Structure

Understanding the file structure helps you navigate and modify the code.

```
PharmaVerse/
├─ package.json          # Dependencies & scripts
├─ vite.config.js        # Build configuration
├─ index.html            # HTML entry point
├─ src/
│   ├─ main.jsx          # React + Router init
│   ├─ App.jsx           # Routes + layout
│   ├─ styles/global.css  # Global styles
│   ├─ platform/         # Platform shell
│   │   ├─ Sidebar.jsx   # Left navigation
│   │   ├─ HomePage.jsx  # Chatbot + use-case cards
│   │   └─ helpers.js    # RAG query + config
│   └─ tools/            # The 4 tools
│       ├─ triallens/     # TrialLens.jsx + data.json
│       ├─ drugnexus/     # DrugNexus.jsx + data.json
│       ├─ bioinsight/    # BioInsight.jsx + data.json
│       └─ fdaforge/     # FDAForge.jsx + data.json
```

3.1 Key Files Explained

- **App.jsx:** The main router. Defines which component loads at each URL path.
- **platform/HomePage.jsx:** Landing page with RAG chatbot and 4 use-case cards. Contains image upload logic.
- **platform/helpers.js:** The `queryPharmaRAG()` function — searches openFDA, then calls Claude API with PubMed MCP.
- **platform/Sidebar.jsx:** Navigation sidebar using React Router's `useNavigate()`.
- **tools/*/data.json:** Embedded datasets. Each tool imports its own JSON.
- **tools/*/*.jsx:** Each tool is a self-contained React component.

4. Tool Deep Dives

4.1 RAG Chatbot (Home Page)

The chatbot uses Retrieval-Augmented Generation — it retrieves real data before generating answers.

How RAG Works

- **Step 1:** User types a question (e.g., 'What is Ozempic?')
- **Step 2:** App extracts keywords and searches openFDA API directly
- **Step 3:** App sends question + FDA context to Claude API, which also searches PubMed via MCP
- **Step 4:** Claude combines FDA + PubMed data into a comprehensive answer with citations

Image Upload

Users upload photos of medicine packaging, pills, or prescriptions. The image is converted to base64 and sent to Claude's Vision API. Claude identifies the medicine and provides information.

4.2 TrialLens — Clinical Trials Intelligence

Route: /triallens | Data: 150+ trials from ClinicalTrials.gov (embedded, CORS blocked)

Module	What It Does
Search	Keyword search with phase/status filters
Analytics	Phase, status, timeline, sponsor, condition charts
Compare	2-3 sponsor comparison with radar chart
Eligibility	Patient-trial matching with scoring
AI Insights	Claude AI chat for trial queries

4.3 DrugNexus — Drug-Target Knowledge Graph

Route: /drugnexus | Data: 20 drugs, 22 targets, 173 diseases from ChEMBL (embedded)

Module	What It Does
Explore	D3.js force graph — drag nodes, click to drill down
Drug Profile	Molecular properties, mechanism of action, indications
Target Profile	Protein info, gene symbols, UniProt links
Pathways	Drug → Target → Disease flow diagram
Compare	Drug comparison with radar chart, shared targets

4.4 BioInsight — Biomedical Literature

Route: /bioinsight | Data: 50 featured + 36M live via PubMed MCP

Module	What It Does
Search	Live PubMed search + featured articles with topic filters
Chat	AI answers grounded in PubMed with PMID citations
Trends	Top journals, article types, PubMed scale by topic

4.5 FDAForge — FDA Approvals Pipeline

Route: /fdaforge | Data: 145 embedded + 29K live via openFDA (CORS supported)

Module	What It Does
Pipeline	Visual ETL diagram + data quality metrics
Approvals	Timeline with year/type/route filters
Analytics	Review priority, dosage forms, pharma classes
Sponsors	Top sponsors all-time vs recent
Explorer	Live openFDA search + embedded browser

5. Key Technical Concepts

5.1 React

JavaScript library for building UIs. Key concepts:

- **Components:** Reusable UI blocks. Each tool is a component.
- **State (useState):** Data that changes — search query, selected tab, messages.
- **useEffect:** Runs code when component mounts or state changes.
- **useMemo:** Caches expensive calculations to avoid re-running every render.

5.2 React Router

Handles navigation without page reloads. URL changes to /triallens but no full page load — React Router just swaps which component is displayed.

5.3 Vite

Fast build tool. Dev mode: hot module replacement (edit → instant browser update). Production: bundles and optimizes everything into a small dist/ folder.

5.4 D3.js

Data visualization library. In DrugNexus, creates force-directed graph where nodes repel each other and links pull connected nodes together. Users can drag nodes.

5.5 Recharts

React charting library. Used for BarChart, PieChart, LineChart, AreaChart, RadarChart throughout the platform.

5.6 Claude API & MCP

Claude is Anthropic's AI. We call it via REST API. MCP (Model Context Protocol) lets Claude connect to PubMed so it can search 36M+ articles in real time.

5.7 RAG

Retrieval-Augmented Generation: retrieve real data first, add it as context, then generate an answer. Makes responses accurate and citeable instead of hallucinated.

5.8 CORS

Browser security rule. openFDA allows cross-origin requests (CORS: *), so browser fetches work. ClinicalTrials.gov doesn't, so we embed that data instead.

6. Deployment

6.1 Build

```
npm run build
```

Creates dist/ folder. Total: ~260KB gzipped.

6.2 Vercel (Recommended — Free)

- Go to vercel.com → sign up with GitHub
- Click 'New Project' → Import PharmaVerse repo
- Vercel auto-detects Vite → click Deploy
- Live at yourproject.vercel.app
- Auto-deploys on every git push

6.3 Netlify (Alternative — Free)

- Go to netlify.com → sign up with GitHub
- Connect repo OR drag dist/ folder
- Build command: `npm run build` | Publish dir: `dist`

6.4 GitHub Pages

```
npm install -D gh-pages
```

Add to scripts: "deploy": "gh-pages -d dist"

```
npm run build && npm run deploy
```

7. Future Development Ideas

7.1 Short-Term (1-2 weeks)

- User authentication (Firebase Auth)
- Save chat history to localStorage
- Dark/light theme toggle
- Mobile responsive design

7.2 Medium-Term (2-4 weeks)

- Node.js + Express backend for API proxying
- PostgreSQL/MongoDB for user data
- Proper RAG with vector embeddings (Pinecone/ChromaDB)
- PDF export for reports
- Drug interaction checker

7.3 Long-Term (1-2 months)

- Python pipeline (Airflow/Prefect) for daily data refresh
- Real-time collaboration
- WHO, EMA API integration
- React Native mobile app

8. Troubleshooting

Issue	Cause	Fix
npm install fails (ENOTFOUND)	Corporate proxy/VPN	Use personal WiFi or hotspot
'vite not found'	Dependencies not installed	Run npm install first
Blank page	JS error	Open DevTools (F12) → Console
Chatbot fails	API not accessible	Works locally with npm run dev
D3 graph blank	DOM not ready	Check console for errors
git push denied	Wrong account	Use token in URL: USER:TOKEN@github.com
Port 3000 in use	Another app on port	Change port in vite.config.js

9. Resume & Portfolio Lines

Ready-to-use bullet points:

- Built PharmaVerse, an integrated pharma data intelligence platform with 4 specialized tools and a RAG-powered chatbot serving real-time data from PubMed (36M+ articles) and openFDA (29K+ drugs)
- Engineered TrialLens, a clinical trials dashboard with search, analytics, sponsor comparison, patient eligibility matching, and Claude AI-powered insights using ClinicalTrials.gov data
- Developed DrugNexus, an interactive drug-target knowledge graph using D3.js force simulation, mapping 20 drugs to 22 protein targets and 173 diseases from ChEMBL
- Created BioInsight, an AI-powered biomedical literature assistant with live PubMed search via Claude API + MCP, paper summarization, and research trend analytics
- Built FDAForge, an ETL pipeline dashboard with live openFDA API, approval trend analytics, review priority analysis, and sponsor intelligence
- Implemented RAG architecture combining openFDA + PubMed MCP + Claude AI for grounded pharmaceutical Q&A with image analysis capabilities

10. Quick Reference Card

What	Command / URL
Install dependencies	npm install
Run dev server	npm run dev
Build for production	npm run build
Dev URL	http://localhost:3000
TrialLens	http://localhost:3000/triallens
DrugNexus	http://localhost:3000/drugnexus
BioInsight	http://localhost:3000/bioinsight
FDAForge	http://localhost:3000/fdaforge
GitHub	github.com/pellurukushalkumarreddy-png/PharmaVerse
openFDA API	api.fda.gov/drug/drugsfda.json
PubMed	pubmed.ncbi.nlm.nih.gov
ChEMBL	www.ebi.ac.uk/chembl
Claude API	docs.anthropic.com
React docs	react.dev
Vite docs	vitejs.dev
Vercel	vercel.com

Good luck with your journey, Kushal! You've built something genuinely impressive. Keep building.